DO NOW - RECALL

Denary to floating point

Answer all four retrieval questions from previous lessons. Use black pen.

Name: _____ Class: _____ Date: _____

1. Choose a type

Which data type is suitable for a measured temperature of 12.75 degrees Celsius?

2. Convert hexadecimal

Convert 3C (base 16) to eight-bit binary and denary.

3. Two’s complement

Represent -26 as an eight-bit two’s-complement integer. Show your method.

4. Explain a fixed-width shift

Logically shift 10110000 one place left in an unsigned byte. Give the stored result and explain why the value is not simply doubled.

DO NOW - CHECK / OBJECTIVES

Correct and improve

Complete page 1 before looking at these model answers. Tick correct work in green; correct mistakes fully.

1. CHOOSE A TYPE

Real / floating point, because the value has a fractional part.

2. CONVERT HEXADECIMAL

3C = 00111100 (binary) = 60 (denary).

3. TWO'S COMPLEMENT

+26 = 00011010; invert to 11100101; add one: 11100110.

4. EXPLAIN A FIXED-WIDTH SHIFT

01100000 (= 96). The leading 1 is discarded; 176 x 2 exceeds the unsigned-byte maximum of 255.

Today: you should be able to...

- Convert denary values, including fractions, to binary.
- Represent normalised floating-point values with an eight-bit two's-complement mantissa and a four-bit two's-complement exponent.
- Normalise a mantissa without altering the represented value.

Key vocabulary

Mantissa — Signed fractional field holding significant binary digits.
Exponent — Signed power of two scaling the mantissa.
Normalisation — Shifting redundant leading sign bits away while preserving the value.
Precision — Detail that can be stored in significant bits.
Range — Span of representable values.

GREEN PEN - Improve one response

Choose a Do Now answer you corrected. Explain the exact misconception and write a stronger answer below.

NEW LEARNING / I DO

Read and annotate

Read the explanations, underline important terminology and add your own examples.

Binary fractions

Binary places after the point are 1/2, 1/4, 1/8, 1/16 and so on. Example: 6.5 is 110.1 in binary; 0.375 is 0.011.

Our OCR teaching format

The value is stored as an eight-bit two's-complement mantissa and a four-bit two's-complement exponent. Value = mantissa x 2^exponent. The binary point is immediately after the mantissa sign bit. No biased exponent is used.

Normalised patterns

A positive normalised nonzero mantissa begins 01. A negative normalised mantissa begins 10. Shift left to remove repeated leading sign bits and decrease the exponent by the number of shifts.

Precision versus range

More mantissa bits improve precision. More exponent bits extend range. Zero uses all-zero fields as a special case.

For this lesson: mantissa bits have weights -1, 1/2, 1/4, 1/8, 1/16, 1/32, 1/64, 1/128. Exponent bits have weights -8, 4, 2, 1.

Explain in your own words

Explain why a positive normalised mantissa starts with 01. Then describe what must happen to the exponent if the mantissa shifts left.

I DO - WORKED EXAMPLE

Represent +6.5 in normalised form

Follow the complete worked example. Each step explains the reasoning, not just the result.

STEP 1 · CONVERT TO BINARY

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6.5 = 110.1 (base 2)
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The whole part is 110; the half is the fractional bit .1.

STEP 2 · MOVE THE BINARY POINT

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110.1 = 0.1101 x 2^3
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Shift three places to the left to form a normalised positive fraction.

STEP 3 · WRITE THE MANTISSA

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01101000
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The eight-bit fraction 0.1101 is padded with zeros on the right.

STEP 4 · WRITE EXPONENT

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0011
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The four-bit two's-complement exponent for +3 is 0011.

Stored pair: 01101000 0011. Check: mantissa 0.8125 x 2^3 = 0.8125 x 8 = 6.5.

Now repeat the method independently

Repeat the method for +5.5, using an eight-bit two's-complement mantissa and a four-bit two's-complement exponent.

WE DO - GUIDED PRACTICE

Try, compare, improve

Attempt each task. Discuss the method before checking with your teacher; show working, not only final answers.

1. Convert a small fraction

Represent +0.15625 using the eight-bit mantissa / four-bit exponent format. Show the binary fraction, normalisation and two fields.

2. Negative value

Represent -6.5 in normalised floating-point form. Start from the positive representation, then negate the mantissa.

3. Normalise a pattern

Normalise 00011010 0101 by shifting the mantissa and adjusting the exponent. Check that the value is preserved.

YOU DO - EXAM PRACTICE

Independent questions · Part 1

Use examination-style working. Do not look at model answers while you attempt these tasks.

1. Positive conversions [4 marks]

Represent (a) 10.5 and (b) 0.375 in normalised floating-point form using eight-bit mantissa and four-bit exponent. Show working and fields.

2. Negative conversions [6 marks]

Represent (a) -10.5 and (b) -0.375 in the same format. Show binary conversion, mantissa negation and final fields.

YOU DO - EXAM PRACTICE

Independent questions · Part 2

Read the command words carefully. Where asked to explain, give the reason and use the correct technical language.

3. Normalise fields [4 marks]

Write 00001011 0111 in normalised floating-point form. Show each shift and exponent adjustment. Adapted from OCR June 2024 Q4(d).

4. Diagnose the mistake [4 marks]

A pupil changes 00101000 0011 into 01010000 0011 and calls it normalisation. Explain why its value changed, correct the exponent and give the correct pattern.

5. Boundary value [2 marks]

Represent -0.5 in normalised form with an eight-bit two's-complement mantissa and four-bit two's-complement exponent.

EXIT TICKET / REFLECTION

Check your understanding

Write T or F and justify any false statement. Show your teacher your answers before checking.

1. Precision

Increasing the number of mantissa bits can improve floating-point precision.

2. Normalisation

Left-shifting the mantissa while leaving the exponent unchanged always preserves value.

3. Patterns

A normalised nonzero negative two's-complement mantissa begins with 10.

Self-assessment

Circle one: Confident / Nearly there / Need more help

GREEN PEN - Correct and explain

Choose one error from your work. Rewrite the correct method and explain why your first answer was incorrect.